BS Projects

Please follow the projects check-list for better project execution, control and delivery.

Display/hide assigned projects

1. Speed/person tracker for home security

Dr. Onaiza Maqbool

Offered: October 2025 *Assigned:* *Student:*

This project is an extension of a previously defined project for home security. The project was related to identifying different types of vehicles from a home security video camera installed in a house, and count their number. As an extension, this project will track the speed of vehicles and also recognize known/unknown people. It will generate events for overspeeding vehicles and unknown people.

2. Salah activity tracker

Dr. Onaiza Maqbool

Offered: October 2025 *Assigned:* *Student:*

Similar to health activity trackers available in smart watches, this app will allow users to track salah and other related activities. Users will be able to define goals and then track their goals. Salah timings will be determined based on user location and time of the year, and the user will be prompted and reminded accordingly.

3. AI-powered EMR Assistant (Through DVISIC Systems Ltd)

Dr. Ayyaz Hussain

Offered: October 2025 *Assigned: October 2025* *Student:*

Build an AI-powered assistant that captures providerâ€"patient audio during consultations, transforms it into structured SOAP notes (Subjective, Objective, Assessment, Plan), integrates with international classification of diseases (ICD10/CPT coding), and syncs seamlessly with existing EMR/EHR systems.

4. AI-based Document Management System (Through DVISIC Systems Ltd)

Dr. Ayyaz Hussain

Offered: October 2025 *Assigned: October 2025* *Student:*

The AI-based Document Management System (DMS) provides an intelligent mobile application that allows users to scan documents using their phone camera, combine multiple scans into a single PDF, or upload documents directly from their device, WhatsApp, Google Drive, or other applications. Users can enhance scanned documents by adjusting contrast, improving readability, or rotating images, and can choose to save them as a single PDF, multiple PDFs, or image files under defined categories and subcategories. An integrated OCR module extracts text from both images and PDFs, storing the results in the database. A machine learning module then preprocesses this text by removing noise, cleaning stop words, extracting key terms, generating metadata, and creating embeddings for efficient document retrieval. Leveraging these embeddings, the system enables smart search and recommendations, allowing users to query in natural language, find and preview the most relevant documents, discover similar documents, or even receive suggestions while drafting new content. The platform ensures quick access, intelligent search, and knowledge-based document drafting, making document management seamless, secure, and AI-powered.

5. Smart Medical Image Annotation Tool

Dr. Ayyaz Hussain

Offered: October 2025 *Assigned: October 2025* *Student:*

Huge number of medical images are generated for different types of diagnosis through imaging modalities like X-rays, CT-Scans, MRI, PET etc. Mechanism is needed to annotate the region of interest (ROI) in medical images for further processing and accurate diagnosis through artificial intelligence-based methods. The purpose of this annotation tool is to initially identify the ROI in a sample medical imaging data after consultation with the radiologist(s). Later, these annotations will be used to automatically annotate the other data of similar nature using semi supervised learning. Tool will provide different options to the radiologists to visualize the annotations and change them accordingly in case of any wrong labelling done by the system.

6. Detection of Clickbait Through Machine Learning

Dr. Muazzam A. Khan Khattak

Offered: October 2025 *Assigned:* *Student:*

There is much research in the detection of malware and malicious traffic using machine learning. The same cannot be said for the detection of clickbait and click bait content. The adoption of social media and the lack of available tools to detect clickbait has meant that individuals are easily falling prey to clickbait. This FYP will explore the creation of a custom solution that explores the detection of clickbait through modern machine learning methods.

7. AI-Powered Feedback System for Student Assignments

Dr. Rabeeh Ayaz Abbasi

Offered: October 2025 *Assigned: October 2025* *Student:*

Giving feedback on student assignments is a time consuming task and often students do not get feedback on their assignments on time. The goal of this project is to develop an AI system that analyzes student assignments using natural language processing and AI to provide constructive feedback. The project will be implemented as a web application. Student taking the project is required to have excellent web development and programming skills and AI knowledge.

8. Agentic static to dynamic website converter

Dr. Rabeeh Ayaz Abbasi

Offered: October 2025 *Assigned: October 2025* *Student:*

Many existing websites have static contents which results into multiple issues like difficulty in updating contents and inconsistency among different pages. The goal of this project is to develop an AI-based solution which would convert a static website into a dynamic website in an interactive mode. The project will understand the existing website and dynamically create the required schema, front-end HTML/CSS/JavaScript code, back-end scripts (PHP), and admin panel for managing the new website. The project will be implemented as a web application. Student taking the project is required to have excellent web development and programming skills and AI knowledge.

9. Archaeological sites virtual 3D recreation using AI

Dr. Khalid Saleem

Offered: October 2025 *Assigned: October 2025* *Student:*

The archeology dept has done numerous budhist historical sites excavation research. The work comprises of text and pictures of excavated sites explaining the site's architectural remains. Artist students add up hand drawn sketches to illustrate the most possible original representation of the sites. The idea is to design and implement a Web based application which takes all the available textual and image data as input and generates a realistic 3D representation of the site using generative AI technology. The development process will be done in close collaboration with the archeology dept of QAU.

10. Vehicle Pollution Testing Home Kit (Mobile Application)

Dr. Khalid Saleem

Offered: October 2025 *Assigned: October 2025* *Student:*

The system takes input for vehicle (Company, model, year etc., different visits to mechanic and maintenance/parts change details). The idea is to design a mobile app, which can collect data of vehicle status at certain date and time. The status data can be Engine Noise (using mobile mic), along with video (using mobile camera); Video can be of belts and other mechanical parts; Silencer exhaust output of different gases and smoke level. This will be achieved using Arduino, which will convert the analogue signals from sensors to digital and forward to mobile app. The app will save all the data in a repository. Mobile app will be equipped with tensor flow lite to predict the maintenance schedule and do different diagnostics of the vehicle, with the help of the input and collected data. The app has to be designed with an easy to use interface, workable on maximum android phones. Secondly, student has to work out the Arduino and sensor working. Already available basic sensors in the market can be used. In the long run, the accumulated data can be integrated with Web available data for better prediction of diagnostics using app level Small Language Model and Web based Large Language Model.

11. Local Weather Monitoring Application (Mobile)

Dr. Khalid Saleem

Offered: October 2025 *Assigned: October 2025* *Student:*

The system takes input of user location, data time, along with environment data (Inside a building, apartment, open area, shade area etc.). The mobile app receives digital data of weather (Temperature, Humidity, Wind, Precipitation etc.), using different sensors, attached to Arduino (for analogue data to digital data conversion). The application has to save data on manual request or at pre-defined time. The app has to be designed with an easy to use interface, workable on maximum android phones. Secondly, student has to work out the Arduino and sensor working. Already available basic sensors in the market can be used. In the long run, the accumulated data can be integrated with Web available data for better prediction of climate using app level Small Language Model and Web based Large Language Model.

12. Time Stamping Crawler

Dr. Khalid Saleem

Offered: October 2025 *Assigned: October 2025* *Student:*

Design and implement a web based application which takes a video as an input. The idea is to time stamp a videoâ€™s first emergence time and date, on the Web. The student has to develop a crawler which can crawl web for same video on the Web, and capture its data and time. An algorithm has to be designed to convert the video into a light weight format (can be intermediate frames in black and white) which can be compared to videos on the web (Tik Tok), in real time. Individuals keep forwarding different videos at different times; the viewer considers it to be latest. The app will be a step forward to manage the original date and time of a video and URL.

13. Price and Quality Comparison Application (PQC)

Dr. Khalid Saleem

Offered: October 2025 *Assigned: October 2025* *Student:*

The idea is to develop a web based information system, which takes an articleâ€™s label and description. And crawls all possible online and physical store articles data (in the userâ€™s region) related to the articleâ€™s description. The application displays the price and quality ratings of the article and similar items. The app should have a very easy to user interface and can correctly process the user request in any language (For testing English and Urdu).

14. AI chatbot for mental health support

Dr. Akmal Saeed Khattak

Offered: October 2025 *Assigned: October 2025* *Student:*

This project aims to develop an intelligent conversational AI chatbot to provide mental health support. Mental health is a worldwide issue. Many people hesitate to seek professional help due to the stigma associated with it. There is a cost and a lack of access to professional support in dealing with mental health issues of concerned individuals. An intelligent conversational chatbot can provide judgment-free support, capable of understanding and responding to mental health concerns. This project aims to interpret usersâ€™ intent, provide personalized self-help content, and detect suicidal thought situations. The preferred technologies include NLP models and libraries, Python (Flask / Django) for backend services, ReactJS for web frontend interface, React Native / Flutter for mobile app, MongoDB / PostgreSQL / Firebase for database management.

15. E-Commerce platform with augmented reality

Dr. Akmal Saeed Khattak

Offered: October 2025 *Assigned: October 2025* *Student:*

This project aims to integrate augmented reality technology with e-commerce platforms to give users a better online shopping experience. The users can virtually try products (clothing, watches, glasses, rugs/carpets, furniture, home decorations, accessories, etc.). This will benefit users in terms of better purchasing decisions. This project provides a user-friendly web and mobile based application benefitting from augmented reality-based visualization tool. This project aims to provide a seamless online shopping experience and support multiple product categories with augmented reality features. The preferred technologies are React.js for web interface, React Native or Flutter for mobile apps, node.js-express/ Django/ flask for backend, MongoDB / PostgreSQL for database.

16. Local food distribution system for hunger relief

Dr. Akmal Saeed Khattak

Offered: October 2025 *Assigned: October 2025* *Student:*

This project aims to develop a technology-driven platform to connect people facing food insecurity, with local food banks and charities gathered through the surplus food from donors (restaurants, supermarkets, caterers, or individual households). This project aims to reduce food waste and ensures the excess edible food reaches the needy people. The problem of food waste and hunger coexist. Edible food is wasted on daily basis and there are individuals and families who suffer from hunger and food insecurity. The preferred technologies are React.js for web interface, React Native or Flutter for mobile apps, node.js-express/ Django/ flask for backend, MongoDB / PostgreSQL for database.

17. AI-Powered Job Recommendation System

Dr. Akmal Saeed Khattak

Offered: October 2025 *Assigned: October 2025* *Student:*

The main objective of this project is match different resumes with relevant jobs. The project will use NLP and collaborative filtering for personalized job recommendations. The student is expected to use embeddings to search for relevant jobs. This project helps students in terms of finding better career and job opportunities. The student is expected to grasp NLP, machine learning, and Flask/Django (web frameworks).

18. Design and Implementation of a Smart POS and Inventory Management System for Renewable Energy Retail Stores

Ms. Ifrah Farrukh Khan

Offered: October 2025 *Assigned: October 2025* *Student:*

Introduction The demand for renewable energy products such as batteries, solar panels, and UPS (Uninterruptible Power Supply) systems has seen significant growth in recent years. Retailers operating in this sector often face challenges in managing stock efficiently, processing sales, and maintaining product-specific information such as warranties and serial numbers. This project aims to address these issues by developing a smart, web-based Point of Sale (POS) and Inventory Management System specifically tailored for such retail environments. Problem Statement Many small and medium-scale retailers in the renewable energy sector rely on manual methods or generic software solutions to manage their sales and inventory. These approaches often result in data inconsistencies, stock mismanagement, lack of analytics, and poor customer service. Additionally, products like batteries and UPS units come with warranty periods and serial numbers, which are often not properly tracked. A customized, centralized system is essential for improving operational efficiency, stock visibility, and customer satisfaction. Objectives The primary objective of this project is to design and develop a robust and user-friendly POS and inventory management system that allows: Efficient processing of sales and returns. Real-time inventory tracking and alerts for low-stock items. Management of product-specific details like brand, capacity, warranty, and serial number. Tracking of purchase history and supplier details. Generation of daily/weekly/monthly reports and analytics for business insights. Secure multi-user access with roles such as Admin, Manager, and Sales Staff.

19. AI-Powered Personalized Diet and Exercise Planner

Dr. Umer Rashid

Offered: October 2025 *Assigned: October 2025* *Student:*

Nowadays, due to a fast-paced and unhealthy lifestyle, it is difficult to manage a proper diet plan. The excessive calories via improper diet intake and lack of proper exercise, the problem of obesity arises that leads to various health-related issues, such as sugar, high blood pressure, heart, Kidney, and liver diseases. The selection of a healthy diet and a relevant exercise/workout plan is cumbersome. In this problem, the student will design and develop an AI-enabled mobile application that mainly approximates the calorie value of the user's diet by taking its snaps. The tool employs advanced image processing and deep learning techniques to approximate the calorie value of the diet. In addition to this, the application recommends personalized healthy diet and exercise plans by considering various attributes of users, diet, and exercise.

20. Personal financial Points Aggregator

Dr. Umer Rashid

Offered: October 2025 *Assigned: October 2025* *Student:*

Nowadays, the excessive use of e-commerce in our daily lives leads to interaction with multiple financial products offered by banks, brands, and shopping outlets etc. The financial products include credit cards, debit cards, insurance policies, and shopping points offered by shopping outlets. The utilization of financial products in daily transactions offers incentives to users, such as discounts, shopping points, and rewards, in a systematic way. However, the perks and benefits are heterogeneous and distributed over multiple financial sites. The lack of awareness and improper access to these perks and benefits negatively impact the userâ€™s financial status. In this project, the student will design and develop an application that can be used as a platform to aggregate the users' benefits from multiple heterogeneous financial sites in a single place. The users will be aware of their financial perks and benefits offered by the various financial sites in a single place in an aggregated way. In this way, the user can gain the maximum benefits from the financial sites in an integrated and usable way.

21. AI-Powered Lead Qualification & Meeting Agent for a Software House

Dr. Umer Rashid

Offered: October 2025 *Assigned: October 2025* *Student:*

Nowadays, a paradigm shift is observed in the design and development of software products. The software houses receive inquiries daily for the development of web, mobile, network, and desktop-based applications from the clients referred to as leads. The extraction of requirements from the leads is complex since they have a non-serious attitude, ask questions without clear intent, are not sure about the software product, etc. The ambiguous requirements, delayed responses, and lack of clients' interest led to improper design and development of software products. The manual calling of the software lead is also cumbersome. In this project, the student will design and develop an AI-Powered Lead Agent that provides a web portal/dashboard to manage and contact the leads. The agent will talk to leads automatically and settle the software design and development-related matters. The agent generates the reports to track the campaign performance.

22. AI-based Fabric Intelligence System

Dr. Umer Rashid

Offered: October 2025 *Assigned: October 2025* *Student:*

Nowadays, the textile industry is very progressive in Pakistan. It is a major source of export. In the textile industry, the production is directly dependent on personal expertise, skills, and the utilization of diverse software tools. The consideration of all these factors in the production of textile products is cumbersome. The negligence of any point may lead to a compromise in the quality of the textile products. In this project, the student will design and develop an artificial intelligence-based system that mainly reduces the manual and fragmented efforts in textile production. The tool may integrate the user experience, expertise, skill, and employment of various tools into a unified production environment. The tool may enhance the textile production performance. In addition to this, the tool reduces the user's efforts and manual activities. The tool may fully automate the textile production process.

23. AI-Based Urdu Handwritten Script Recognizer's Educational Assistant

Dr. Mudassar Azam Sindhu

Offered: October 2025 *Assigned:* *Student:*

This project requires to develop a web and Android application that uses machine learning to recognize handwritten Urdu characters, words, or short sentences from images (e.g., scanned notes or mobile camera input or digitally written). The developed system will aid conversion of handwritten text into editable digital Urdu (Nastaleeq script) and provide grammar suggestions, synonyms, and translation to English. This will provide aid to student preparing for Urdu text (essays/paragraphs) along with training them for translation to English text.

24. Automated Exam Malpractice Detection Using Computer Vision

Dr. Mudassar Azam Sindhu

Offered: October 2025 *Assigned:* *Student:*

This project requires to build a web-based remote/online monitoring tool employing computer vision to detect potential malpractice during online exams. It will use a student's webcam feed and will analyze his/her behavior in real-time: by detecting multiple faces, phone usage, looking around frequently, or unauthorized objects close by. Alerts will be logged and displayed on the invigilatorâ€™s dashboard for review. The project can also be integrated with an existing LMS of the university.

25. Fault Prediction's Test Case Prioritization Tool for Legacy Software

Dr. Mudassar Azam Sindhu

Offered: October 2025 *Assigned:* *Student:*

This project requires creating a web-based intelligent software testing assistant that will predict high-risk modules in legacy codebases and will prioritize test cases based on historical bug reports and code complexity metrics (e.g., cyclomatic complexity, LOC, FPs, churn rate). It will be Integrated with GitHub/GitLab to analyze commit history and suggest where developers should focus regression testing. This will help Pakistani IT companies in reducing effort for maintaining large legacy systems. This tool will improve QA efficiency and will reduce release-cycle time.

26. AI-Powered Chat-Based Academic Advisor for CS Department (Web-Based Intelligent Assistant)

Dr. Mudassar Azam Sindhu

Offered: October 2025 *Assigned:* *Student:*

This project requires to develop an intelligent chatbot system that serves as a 24/7 virtual academic advisor for students of the Computer Science Department at QAU. Hosted on the departmental intranet or integrated into the official QAU-CS website, this AI-powered assistant will allow BS, MS/MPhil, and PhD students to ask natural language queries about: i. Course registration procedures and deadlines ii. Credit hour requirements per semester iii. Prerequisites for specific CS courses (e.g., "Can I register Machine Learning without Data Structures?") iv. Fee structure based on program (BS/MS), shift (morning/evening), and number of registered courses v. Timetable information (class schedules, lab timings, exam dates) vi. Thesis submission guidelines, course exemption policies, and academic progression rules The developed system will use Natural Language Processing (NLP) techniques to understand user intent, extract entities (like course codes or student type), and retrieve accurate responses from a structured knowledge base derived from the CS departmentâ€™s academic regulations, syllabi, and administrative policies. Students / users can interact via a web-based chat interface, with support for both English and basic Urdu input using a web browser on their Windows/Mobile devices. The backend will make use of a rule engine employing a fine-tuned lightweight language model (e.g., BERT-based Urdu/English multilingual model or DistilBERT) for improved understanding of academic terms.

27. AI-Based Personal Posture & Ergonomics Coach

Dr. S. M. Naqi

Offered: October 2025 *Assigned: October 2025* *Student:*

This project aims to design and develop an AI-enabled application that assists individuals in maintaining proper posture during long working or study hours by applying computer vision techniques. The system will continuously monitor the userâ€™s posture through a laptop or mobile camera, detect key skeletal points such as the neck, shoulders, and spine, and classify the posture into categories like â€œgood postureâ€□ or â€œbad postureâ€□ (slouching, leaning, forward head, etc.). When poor posture is detected for a prolonged period, the app will provide real-time feedback in the form of notifications, sounds, or visual alerts, helping the user correct their position immediately. The system will also maintain a log, and on the basis of these statistics, weekly/monthly reports will be generated. The system will also give recommendations for ergonomic improvement. This project will address the posture-related issues that are becoming increasingly common among students, office workers, and remote employees.

28. Real-Time Deepfake Detection in Live Video Calls

Dr. S. M. Naqi

Offered: October 2025 *Assigned: October 2025* *Student:*

With the rapid rise of online meetings, interviews, and teleconferencing, malicious actors are increasingly using deepfake technology to impersonate individuals in real time, posing serious risks to organizational security, financial transactions, and personal privacy. Most existing deepfake detection systems are designed for pre-recorded content and are not optimized for live streaming environments. This project aims to develop a lightweight, real-time deepfake detection system that integrates directly with video conferencing platforms. The proposed system will analyze facial landmarks, blink rates, head movements, and temporal inconsistencies to detect potential deepfakes during live calls. By applying a combination of computer vision and temporal analysis techniques, the model will process video frames in real time and provide

authenticity scores or instant alerts when suspicious activity is detected. Such a tool can be valuable for corporate security, online education, legal hearings, and government institutions by safeguarding virtual interactions and maintaining trust in digital communication platforms.

0/28 projects assigned. 28 projects unassigned.

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